

Client

(Network)

Server

Classical Phase

ClientHello [TLSv1.3 + ECDH-256 key_share]

recv

ServerHello [ECDH-256 key_share]

recv

Compute ECDH:
 SS_{ECDH}

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Classical security
(ECDH-256)

PQC Extension: ML-KEM-768 pk_client

Encapsulate:
 $(ct, SS_{ML-KEM}) \leftarrow$
 $KEM.Encap(pk_{client})$

PQC Phase

PQC Extension: ML-KEM-768 ciphertext ct

Decapsulate:
 $SS_{ML-KEM} \leftarrow$
 $KEM.Decap(sk, ct)$

Post-quantum
security (ML-
KEM)

Key

Combine Shared Secrets:
 $SS =$
 $KDF(SS_{ECDH} || SS_{ML-KEM})$
TLS 1.3 Key Schedule:
→ handshake_traffic_secret
→ application_traffic_secret

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Hybrid key
schedule (KDF)

Auth Phase

Certificate [X.509 + ML-DSA-65 pubkey]

CertificateVerify [ML-DSA-65 sig + ECDSA-256 sig]

Finished [HMAC-SHA256]

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Security guarantee: Confidentiality holds if either ECDH-256 or ML-KEM-768 is secure.
Authentication uses composite signatures (ML-DSA-65 + ECDSA-256) for backward compatibility.